

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT1-Assignment

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QUESTIONS

Q1

Banking Transaction System –
Exception Handling
10 marks

Q2

Hospital Patient Registration –
User-Defined Exception
10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions
10 marks

Q4

Online Food Delivery – Multiple
Threads and Execution Order
10 marks

Q5

Manufacturing Plant –
Producer-Consumer Problem
10 marks

5/5 answered

Q3

**E-Commerce Order Processing – Built-in
Exceptions**10
marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 `ArrayIndexOutOfBoundsException` handled

Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 `InputMismatchException` handled Invalid input

Your Code

```
1 import java.util.*;
2
3 public class Main{
4     public static void main(String[] args){
5         String[] prod = {"Laptop","Phone","H
6         double[] prices = {1000.0,500.0,100.
7
8         Scanner sc = new Scanner(System.in);
9
10        while(sc.hasNext()){
11            try{
12                int index = sc.nextInt();
13                int quantity = sc.nextInt();
14
15                if(index < 0 || index >= pro
16                    throw new ArrayIndexOutO
17            }
18
19            if(quantity <= 0){
20                throw new IllegalArgumen
21            }
22
23            double totalCost = prices[in
24            System.out.println("Product:
25            System.out.println("Valid pr
26        }catch(InputMismatchException e)
27            System.out.println("InputMis
28            sc.nextLine();
29        }catch(ArrayIndexOutOfBoundsExceptionExce
30            System.out.println(e.getMess
31        }catch(IllegalArgumentException
32            System.out.println(e.getMess
33        }
34    }
35    sc.close();
36 }
37 }
```



Input

1
2

Output

```
Product: Phone, Total Cost: 1000.0  
Valid product, total cost calculated Normal  
case
```

✓ Submitted

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